PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila











PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

NUTRIFEED

Gut Health Optimization:

The Role of Dairy-Derived Ingredients in Pre- and Post-Weaning Piglet Diets

Mr. J. Lemmen
Nutrifeed Animal Nutrition

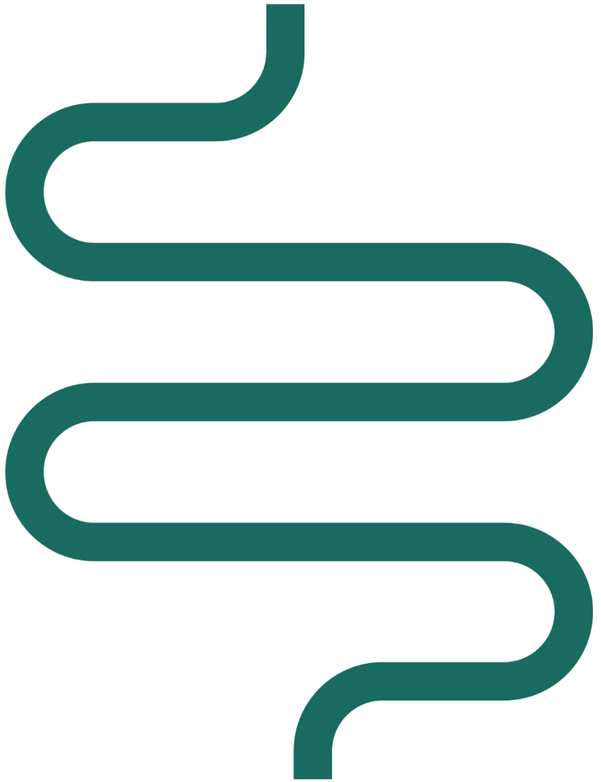
Importance of Strengthening Young Animals' Gut Health

A healthy gut is crucial for health and growth.

The gut is only 5-7% of the body mass but represents:

- 20-35% of the whole body energy consumption
- 20-60% of the protein requirement
- 70-85% of the immune system

In this way, the gut has a big effect on the technical performance of the piglet!

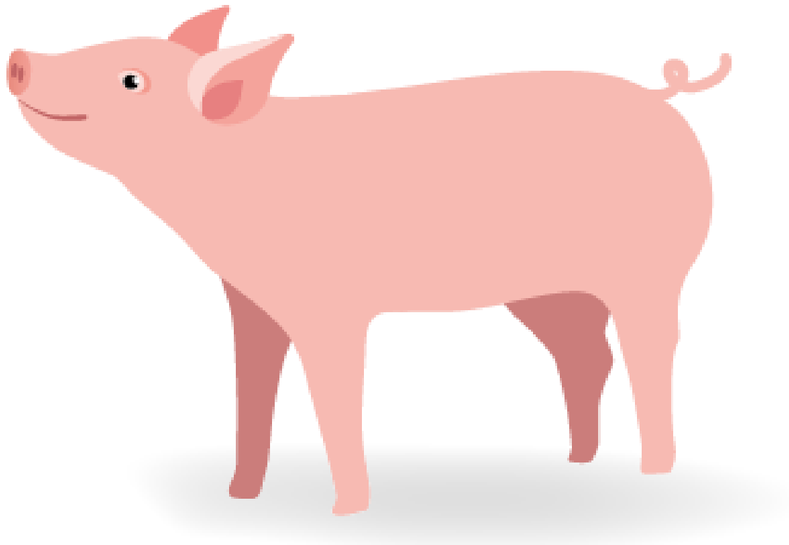


Post-weaning challenge

A piglet is not born as an eater!

What causes post-weaning health problems?

- Reduced feed intake in the first day's post-weaning
- Abrupt switch from sow milk to dry feed
  - Change of microbiota
- Enzyme production is immature to digest nutrients provided at weaning
- Inability to effectively acidify the stomach sufficiently
  - Increased amount of undigested protein entering the intestinal tract

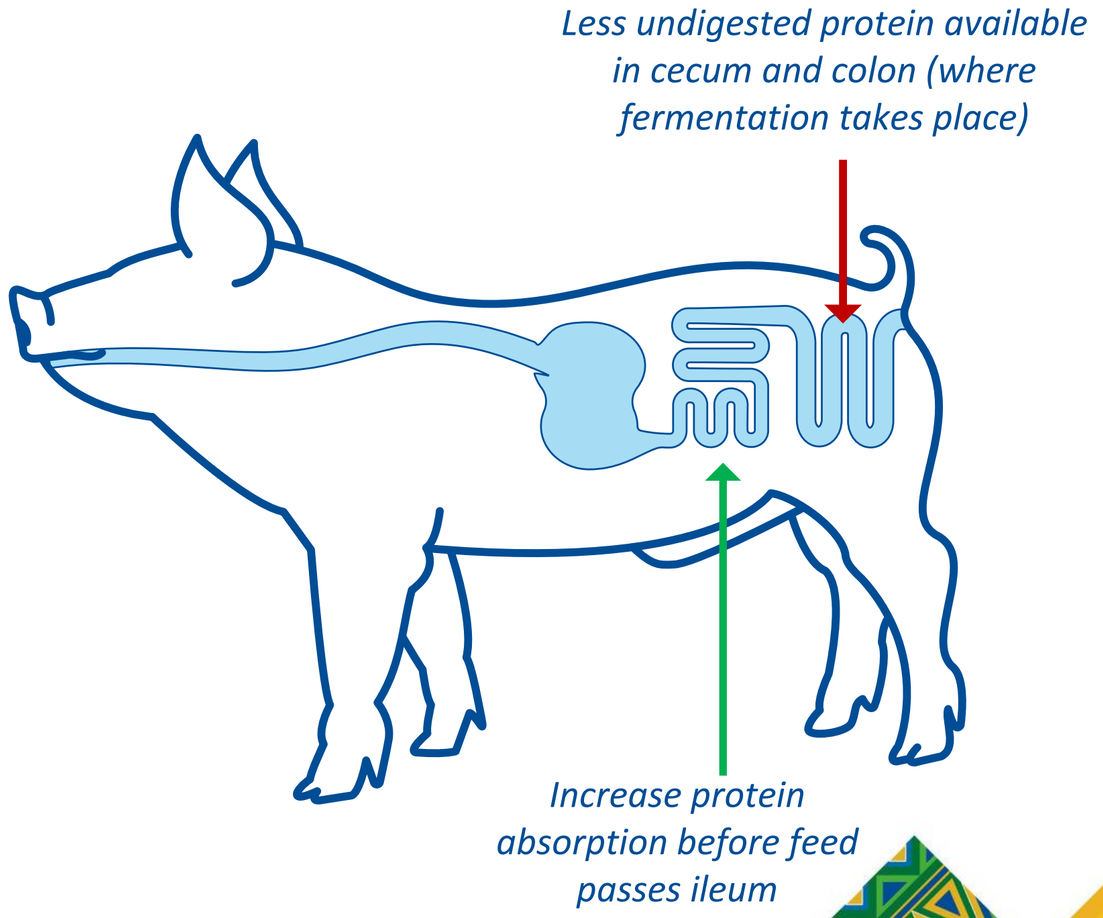


What if protein digestion is not optimal?

1. **Undigested protein** can be used as **a substrate for fermentation** by **pathogenic bacteria**.

2. Pathogenic bacteria can **produce toxic metabolites** and **rapidly increase in number** (proliferation).

3. This **harms gut health** and could **cause diarrhoea**.



Importance of a good (dairy) protein source

The **digestive capacity** of a piglet **can be optimized** and **stimulated** to adapt to digest nutrients.

e.g., gradual switch from dairy protein to vegetable protein

This will avoid large quantities of undigested protein available for fermentation by pathogenic bacteria having a negative effect on gut health.

As the piglet is not used to digest vegetable protein sources, **highly digestible (dairy) protein sources are needed to support the piglet.**

Dairy protein has the highest digestibility in just weaned piglets



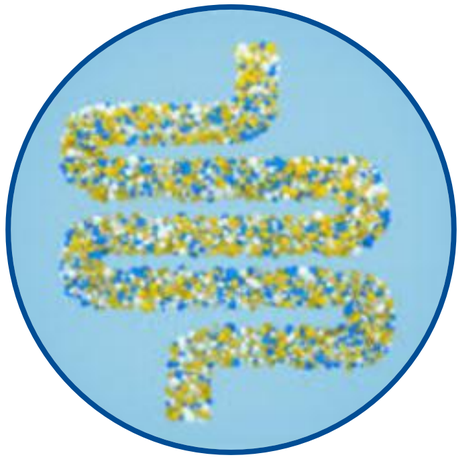
How can dairy help young piglets?



The right nutrients

Dairy ingredients are just like sow milk packed with the right nutrients, which are very palatable and easy to digest.

Providing essential building blocks and energy.



Piglet health

Dairy ingredients can support the development of a healthy gut and immune system.



Improved performance

The use of dairy products supports improved performance, lower mortality, and profitable piglet rearing.

Dairy ingredients supporting gut health

Dairy acts as a nutritional bridge: supporting gut maturation and resilience

Lactose: an easy available energy source.

Highly digestible **milk proteins (whey and casein)**, adapted to an immature digestive tract.

Bioactive compounds to support immune function.

- Lactoferrin
- Lactoperoxidase

Milk oligosaccharides promote the growth of beneficial bacteria.

Constituents of cow's milk



dry matter

- Milk fat (32%)
- Casein (20%)
- Whey protein (6%)
- Lactose (36%)
- Ash (6%)


Butter



Insoluble part of milk

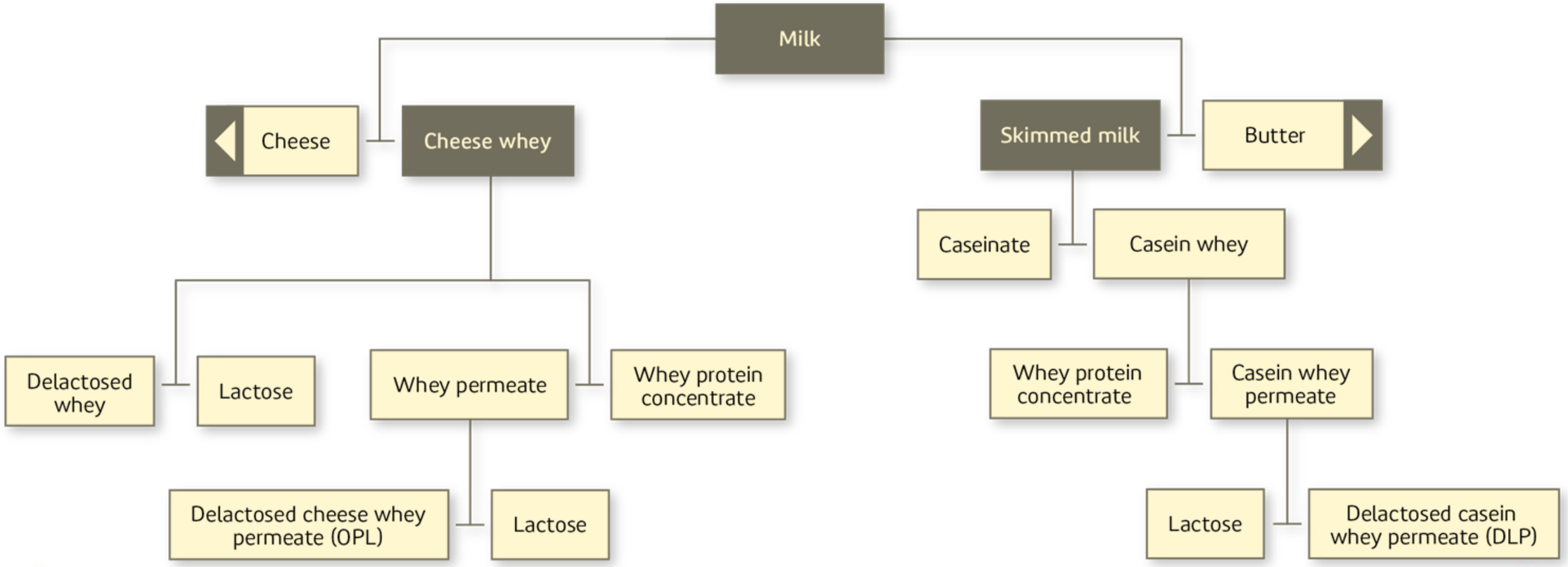

Cheese



Soluble part of milk

Dairy streams

Single ingredients derived from milk processing may differ in quality and composition



Various dairy products are available

	Protein	NPN	Ash	Fat	Moisture	Lactose	pH
	%	% of N	%	%	%	%	
Skimmed milk powder	35	6	8	0.7	3-4.5	50	6.00-6.60
Sweet whey	11-13	20-30	7.5-8.5	1-2	1.5-4.5	70	6.00-6.60
Lactic acid whey	8-10.5	50-60	10-13	1-2	1.5-4.5	60	4.50-5.10
Acid whey	9-11	25-30	10-12	<1	1.5-4.5	70	5.60-6.60
Demineralised whey	12-14	20-30	5-6	1-2	1.5-4.5	72	6.00-6.60
Delactosed whey	25-27	25-30	15-17	2-3	3-4.5	45-48	6.00-6.60
WPC	30-75	5-15	6-8	3-15	3.5-5.5	18-50	6.00-6.60
Whey permeate	0-4	>80	8-8.5	<1	1.5-4.5	80-85	6.00-7.00
Casein	90	0	<1	<1	6-8	0	4.30-5.00
Caseinate	< 88	0	2-3	<1	2-4	0	6.00-7.00

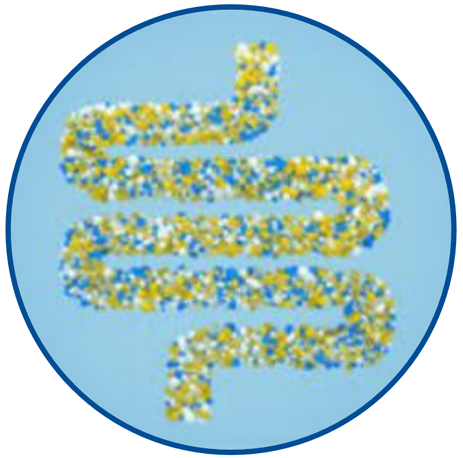
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Improved performance

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Sow milk

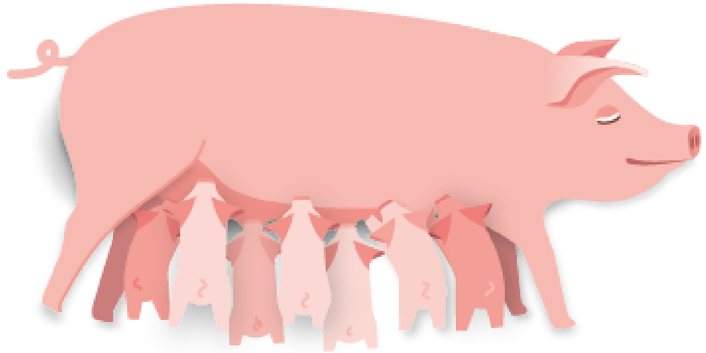
The most important source of nutrients

Sow milk is adapted to the needs of the young piglet:

- Antibodies (colostrum): to protect against pathogens

- Energy: piglets have limited energy reserves and have difficulty maintaining body temperature
  - **Lactose:** the most important energy source
  - **Fat:** energy and thermoregulation
  - **Proteins:** growth and development

- Bioactive components: to stimulate gut development and maturation of the immune system



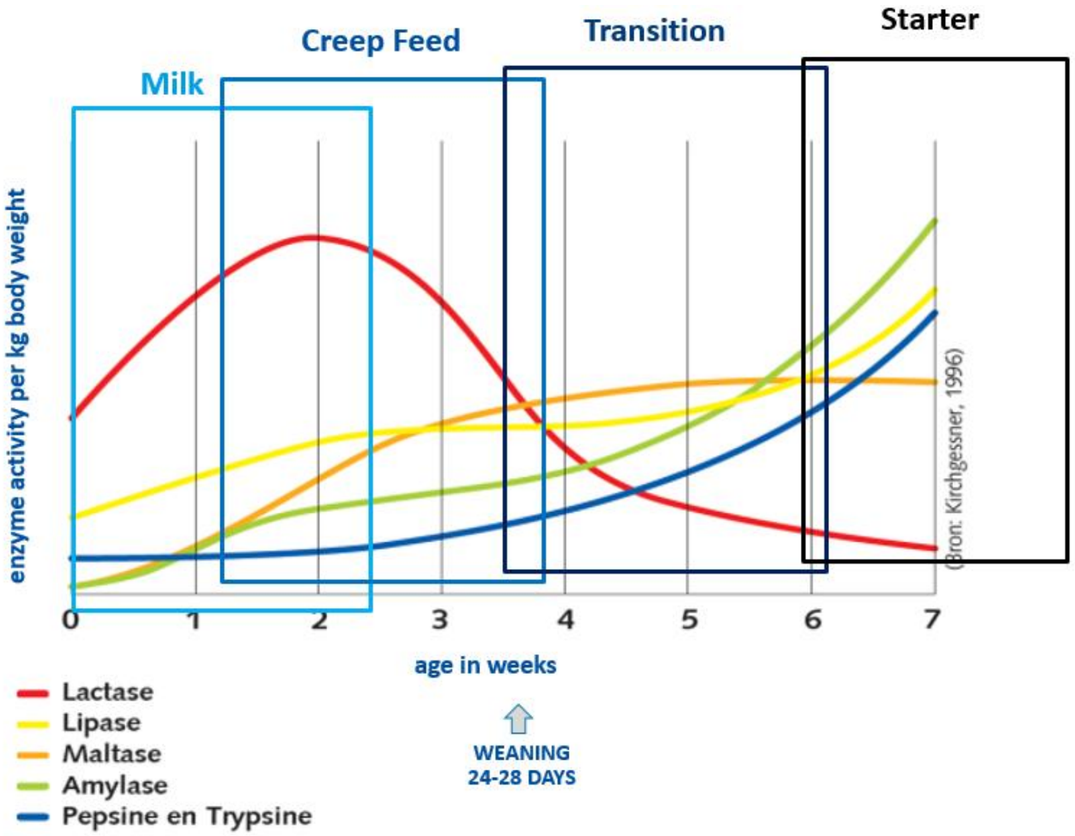
The weaning challenge

Piglets' enzyme production is not capable of effectively digesting the nutrients provided at weaning

- Piglets are adapted to digest sow's milk
- Digestive enzyme production is still immature: nutrient digestion may be compromised

How dairy can help

- Provides highly digestible nutrients that match the capacity of the young piglet
- Facilitates a smoother transition to plant-based feeding



Digestibility of dairy protein

In a trial performed by Schothorst Feed Research, dairy protein (from skimmed milk powder) was found to have the highest digestibility in just-weaned piglets, compared to vegetable sources.

Raw material	Apparent faecal crude protein digestibility coefficient (%)
Fish meal	82.6^{c}
Potato protein	91.1^{de}
Skimmed milk powder	**94.5^{e}**
SPC	87.1^{cd}
Soya bean meal >48% CP	82.3^{c}
Rapeseed meal	66.5^{a}
Sunflower seed meal hipro	74.1^{b}
Maize gluten	91.5^{de}

Source: Schothorst Feed Research report no. 1696, 2019

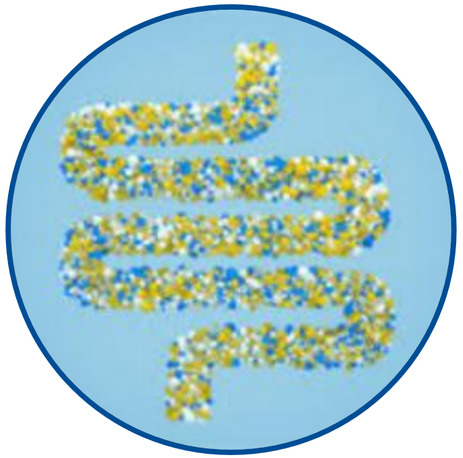
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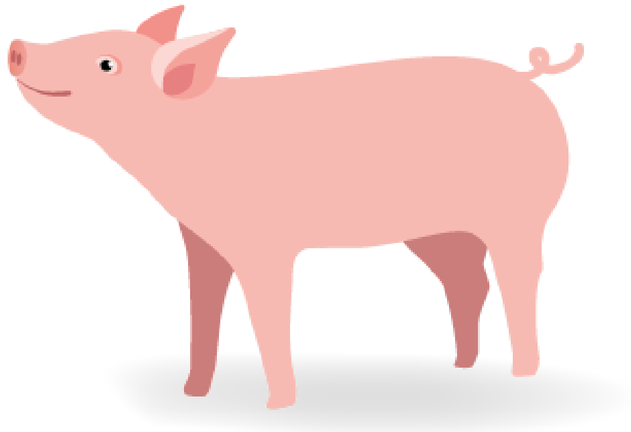
Improved performance

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Comparing protein sources post-weaning

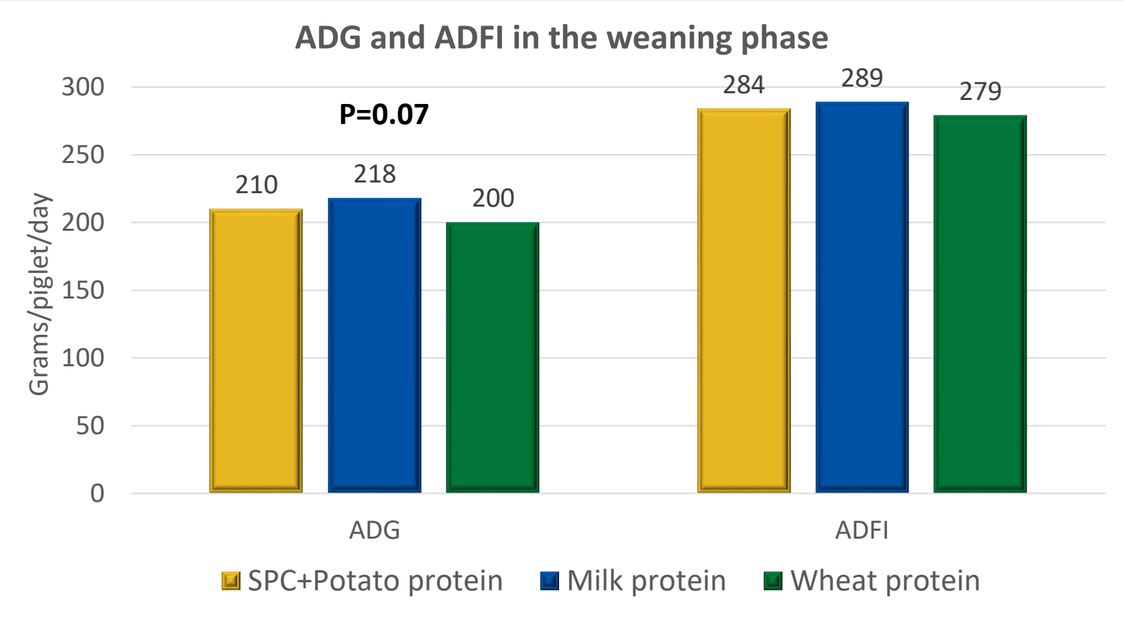
Dairy protein vs. SPC/potato protein vs. wheat protein

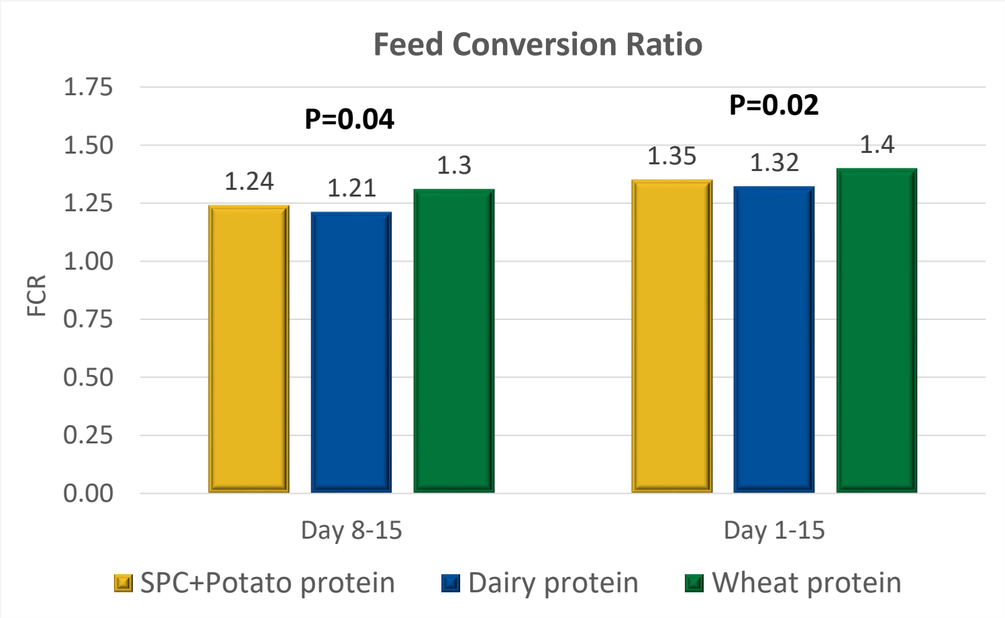
- Weaning at 27 days with an average weaning weight of 9.0 kg
- Average (24 pens) per treatment
- TN70 x Tempo piglets
- Weaner diets days 0 – 14 after weaning
  - Control (SPC + potato protein)
  - 3% Milk protein (WPC 35)
  - 2.8% Wheat Protein
- All 3 groups received the same rearing diet from 15-36 days post-weaning



Results

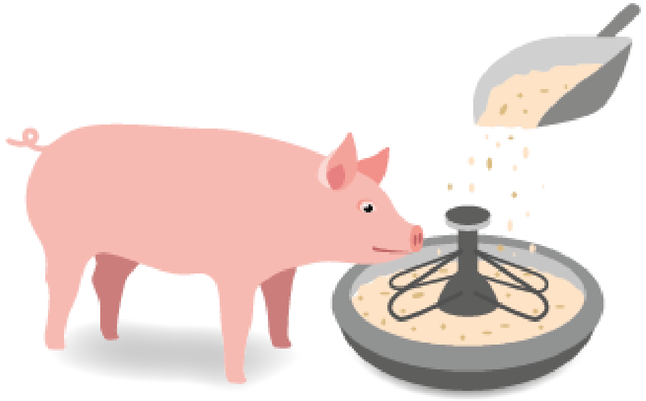
Milk protein as the main protein source in weaning diets led to the highest performance!





A piglet is not born as an eater!

A better pre-weaning preparation could increase performance post-weaning



Facilitate maximum growth potential of piglets

- Sow milk alone is not sufficient to achieve maximum growth
- Extra energy for piglets
- Reduce mortality

Assisting the sow by supplementing piglet milk and creep feed additionally to sow milk

- Compensation for shortcomings or absence of the sow
- Has a positive effect on GIT health and development
- Piglets are better adapted to digest (vegetable-based) post-weaning diets

The benefit of using dairy protein in a creep feed

A healthy gut begins with a strong start!

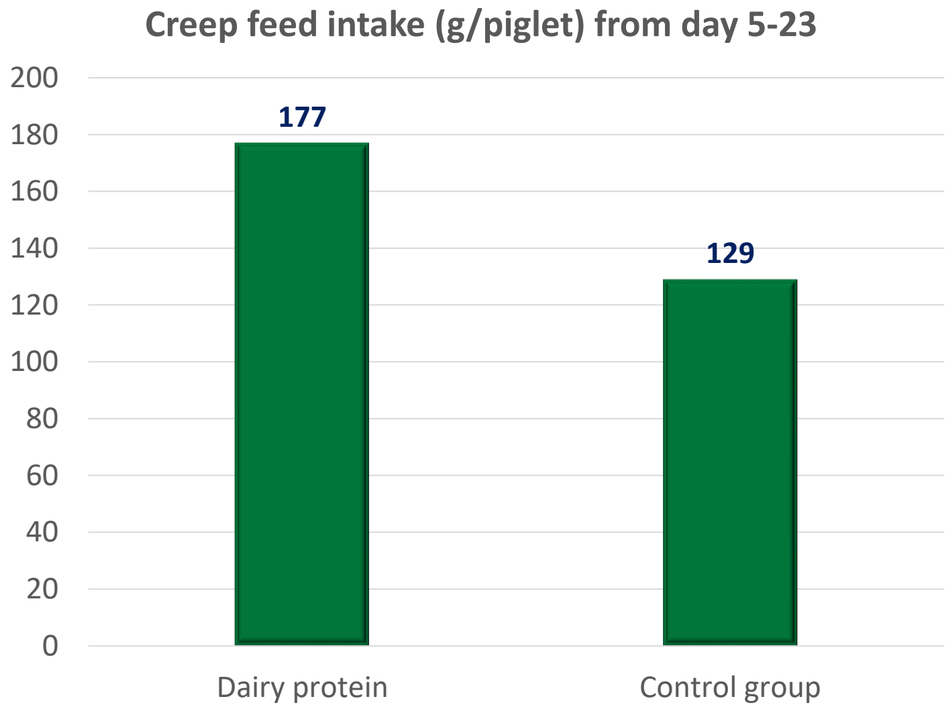
A creep feed with milk protein results in a higher intake from day 5-23!

Follow-up post-weaning

The creep feed from pre-weaning was provided for 4 days post-weaning:

Improved faecal scores were seen in the dairy protein group:

- ✓A higher number of pens with dark grey and black faeces
- ✓A higher number of pens with firm faeces, indicating better digestion of the diet



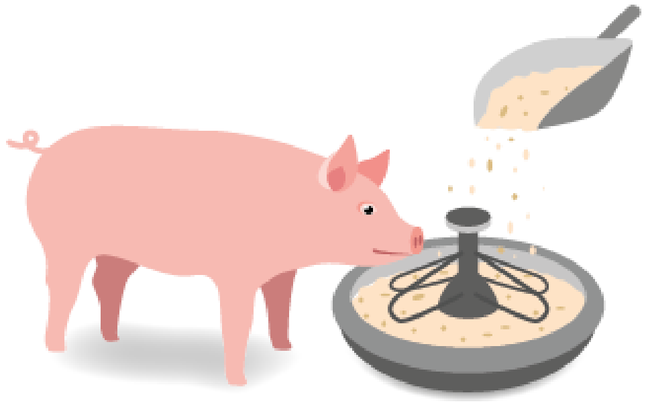
Impact of lactose nutrient level in creep feed

- Creep feed with the same dairy protein and energy levels

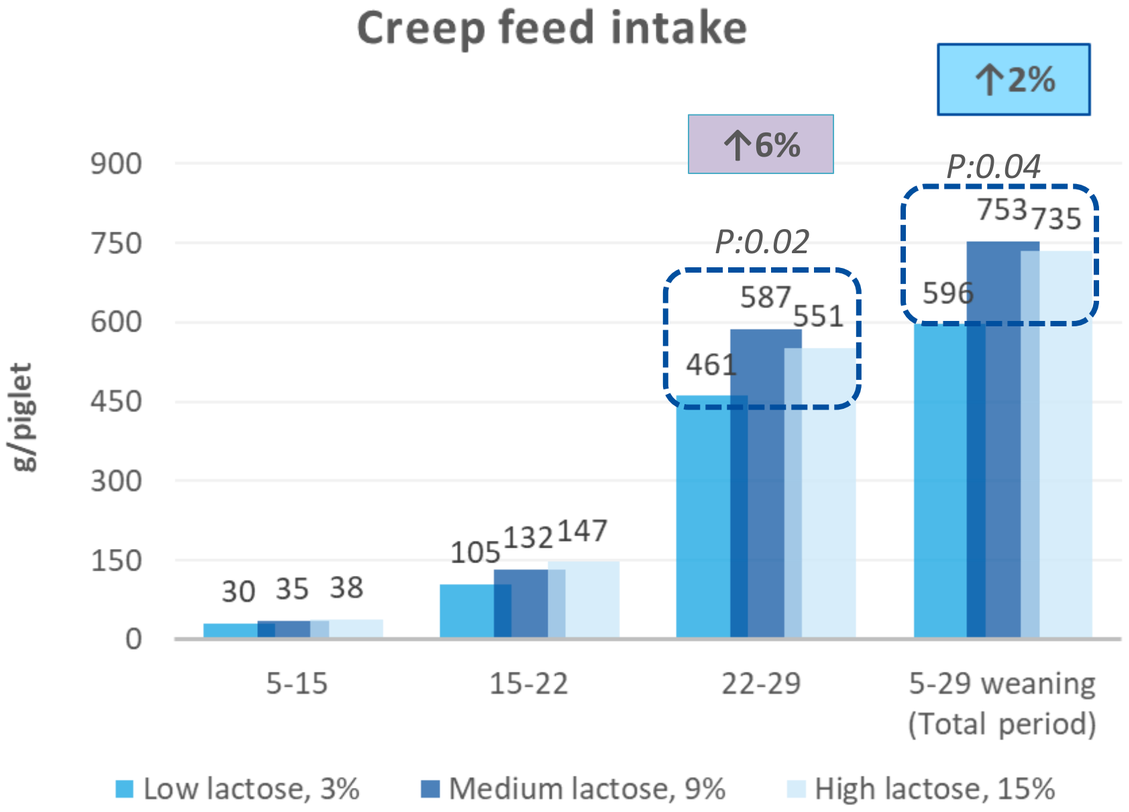
Potential effects:

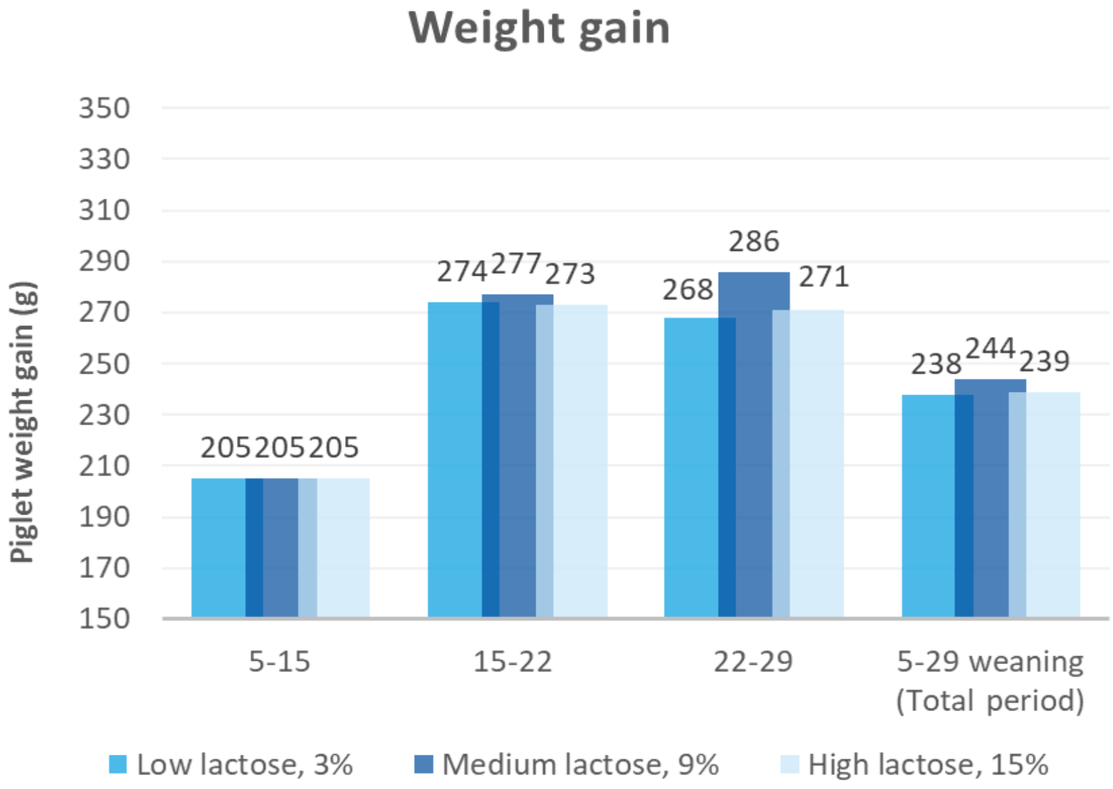
  - ↑ Creep feed intake
  - Laxative effect?
- No control group
- Genetics: Topigs TN70 x PIC Pietrain

Number of pens	A	B	C
Day 5-29 (age 3 days – weaning)	Low-(3%) 2 mm micropellet (dry)	Medium-(9%) 2 mm micropellet (dry)	High-(15%) 2 mm micropellet (dry)



Impact of lactose nutrient level in creep feed



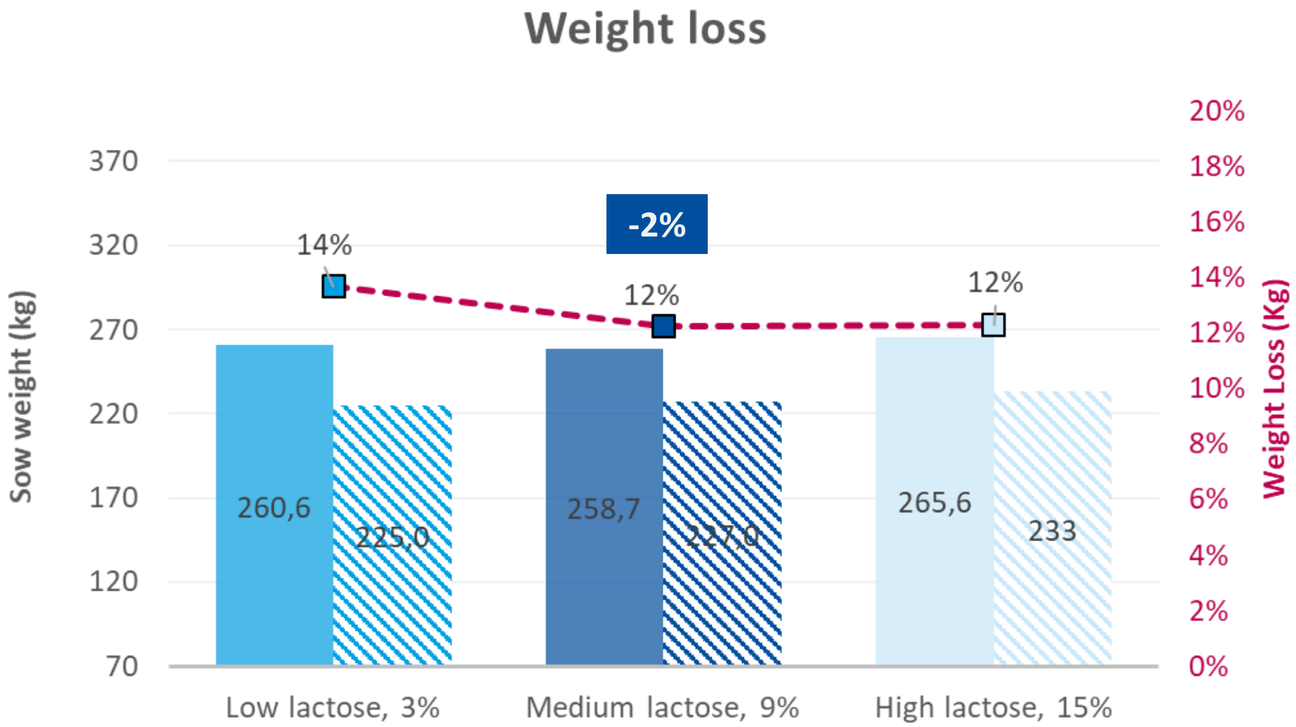


The level of lactose **significantly increased** creep feed intake. There was no significant difference in the incidence of abnormal faeces.

Although the feed intake increased, the growth rates appeared similar. Possibly due to an increased demand for sow's milk.

Impact of lactose nutrient level in creep feed

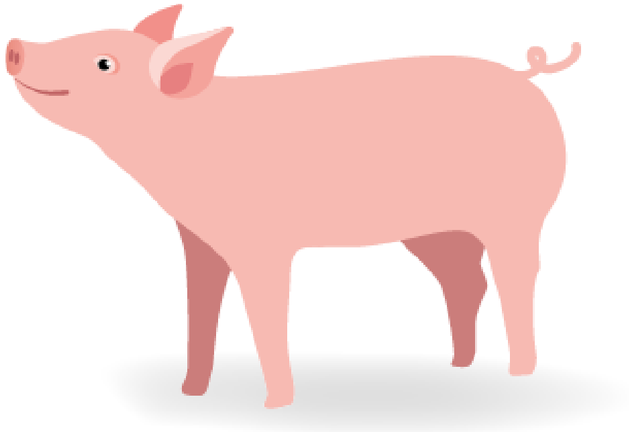
- The **highest weight loss** was observed in the group of piglets that had the **lowest lactose level** and creep feed intake
- **Increased lactose levels** lead to **greater feed intake in piglets**, which reduces the demand on the sow
- Piglets that struggle to adapt to dry feed will demand more milk from the sow.
- Sows that lose more than 12% of their body weight tend to have longer intervals between weaning and service, resulting in reduced reproductive rates.[1]



[1] *Louisa Zak - Global Reproduction Platform Lead at Topigs Norsvin / WIAS Symposium – Nutritional Management of Lactating Sows, September 2024*

Take home message

- **Supporting gut health is crucial** for **growth** and **resilience post-weaning**
- **Dairy contains the right nutrients** that match the piglet's digestibility capacity to smooth the transition from sow milk to plant-based feed
- Dairy ingredients may **differ in quality** and **composition**
- Providing dairy protein **significantly increases performance post-weaning**
- A piglet is not born as an eater: **supporting gut health starts already in early life**



A smoother transition starts with nutrients the gut already knows!

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Thank you for your attention



Mr. J. Lemmen
Nutrifeed Animal Nutrition

E info@Nutrifeed.com
WWW.NUTRIFEED.COM

Partner in the Philippines, **Diconex Phils., Inc.**